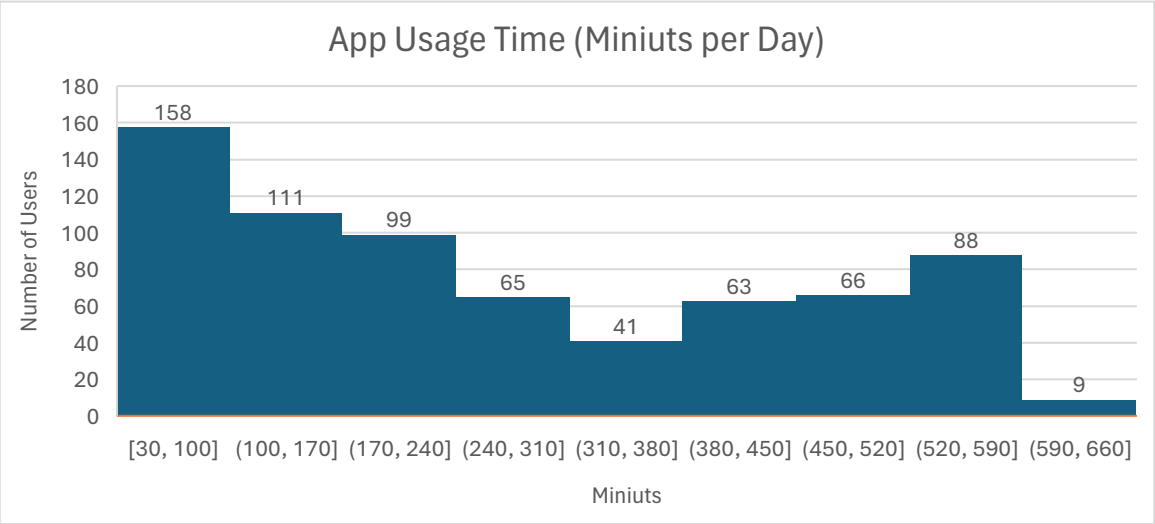
Objectives/Questions with Anlysis

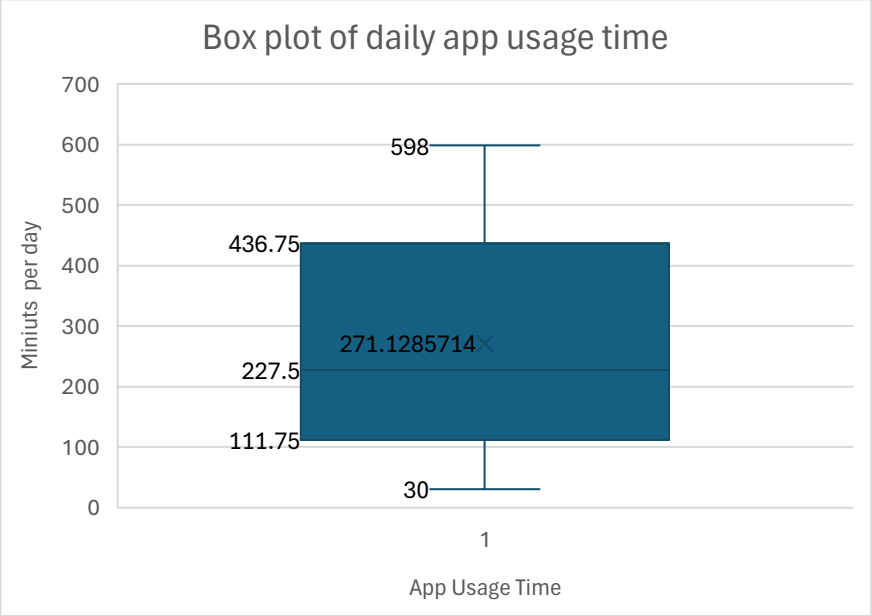
1. **How much time do users spend on apps per day, and how is that time spread across users?**

Mean	271.1285714
Median	227.5
Min	30
Max	598
Q1	113.25
Q3	434.25
IQR	321
Std Dev	177.0728681

- The **mean app usage time** is **271 minutes per day**, while the **median** is **227.5 minutes**, showing that typical users spend around **3–4 hours daily** on apps.
- The **minimum** usage is **30 minutes**, and the **maximum** is **598 minutes**, indicating a wide range of user behavior.
- The **interquartile range (321 minutes)** shows that app usage varies greatly among users.



- The histogram shows that **most users spend low to medium amounts of time** on apps per day.
- As app usage time increases, the number of users gradually decreases.
- A small number of users spend very high amounts of time on apps, creating a **right-skewed distribution**.

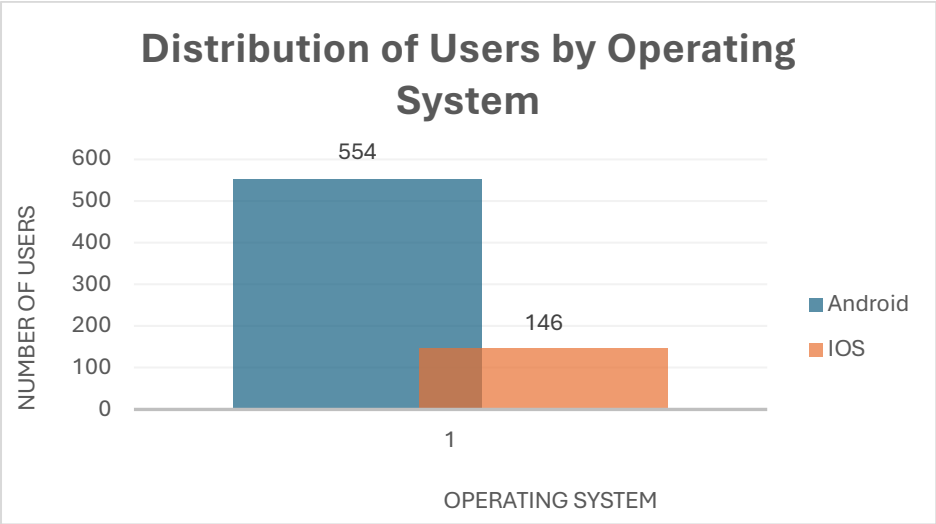


- The box plot shows that the **median app usage** lies closer to the lower part of the data range.
- The wide box and long upper whisker indicate **high variability** in usage time.
- A few users appear to be **heavy app users**, spending significantly more time than most users.

Verdict: Most users spend a moderate amount of time on apps each day, while a few users spend very high amounts of time. App usage time varies widely and is right-skewed due to these heavy users.

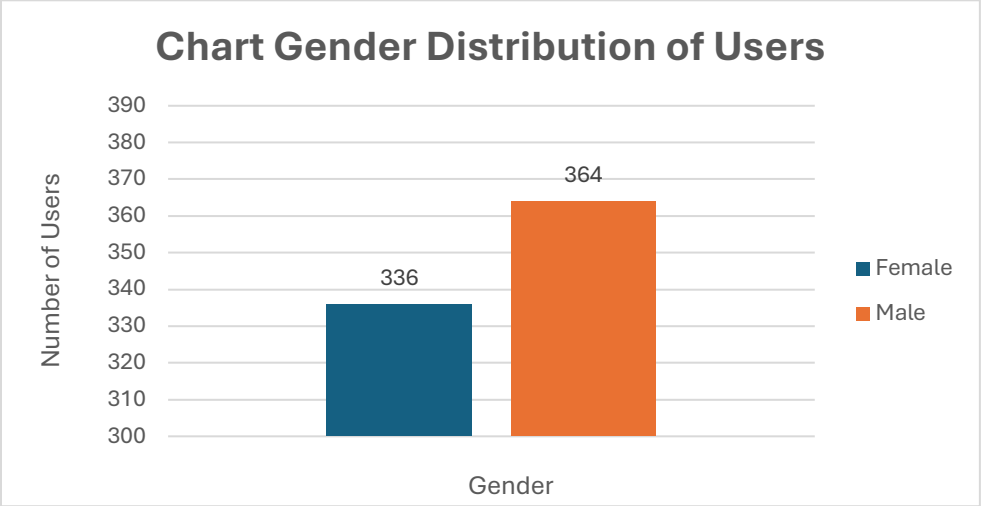
2. How many users belong to each category and what percentage do they represent?

i. Android vs IOS users



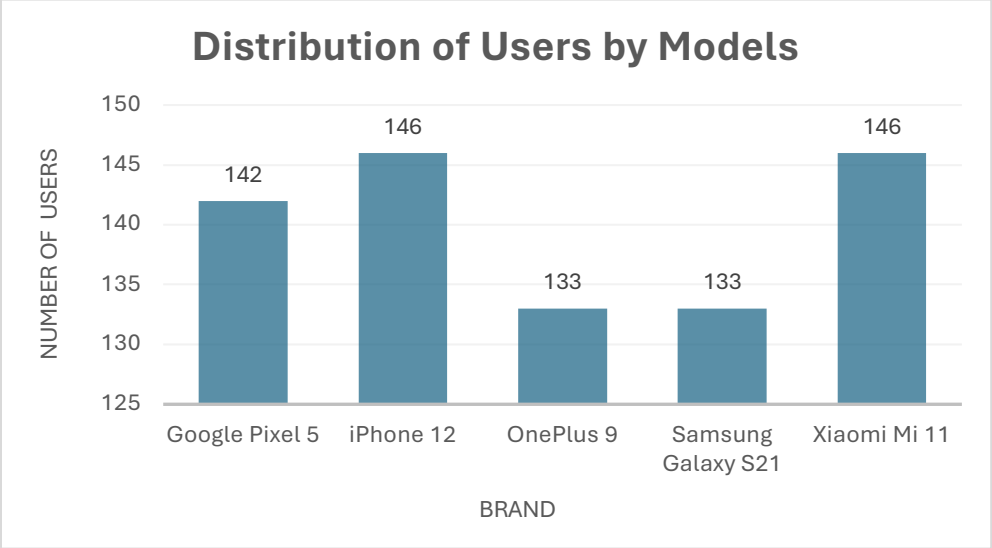
The distribution shows that Android users form the majority of the dataset, representing 79.14% (554 users), while iOS users account for 20.86% (146 users). This indicates a moderate dominance of Android users within the sample.

ii. Male vs Female distributions



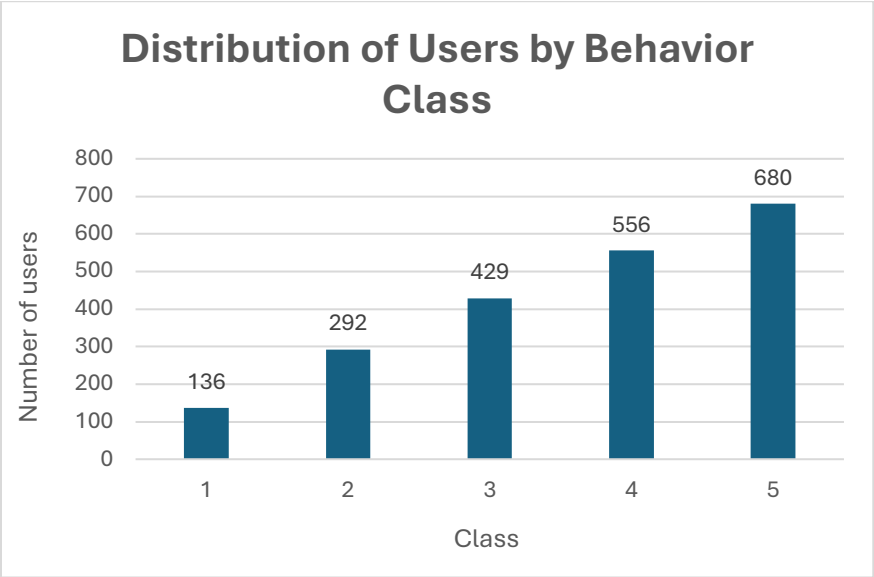
The gender distribution appears relatively balanced, with males accounting for 52% and females 48%. This suggests that the dataset does not exhibit a strong gender imbalance.

iii. Most common models



The models distribution reveals that iPhone 12 and Xiaomi Mi 11 are the most used models, representing 20.86% of the dataset. In contrast, Samsung Galaxy S21, one plus 9 are the smallest share. This indicates that user representation is concentrated among a few dominant models, while others appear less common.

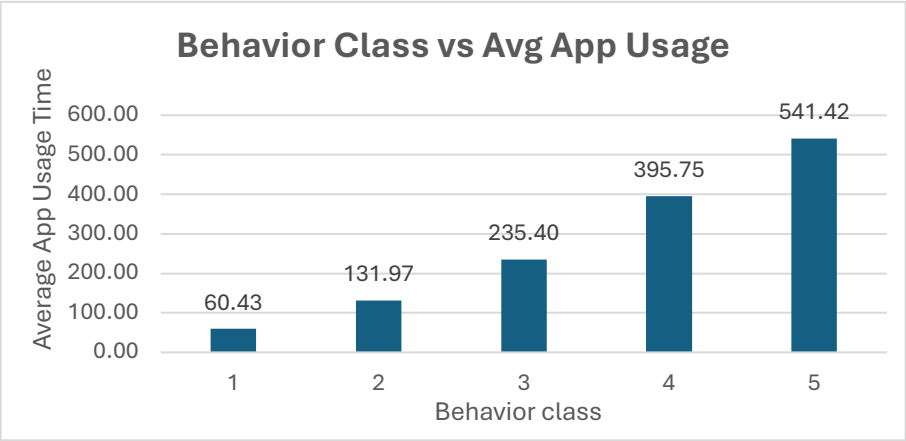
iv. **How many users in each behavior class (1–5)**



The distribution across behavior classes shows that Class 5 has the highest number of users, accounting for 32.49 % of the dataset. Class 1 has the lowest representation. This suggests that most users tend to fall within the 6.50 behavior category

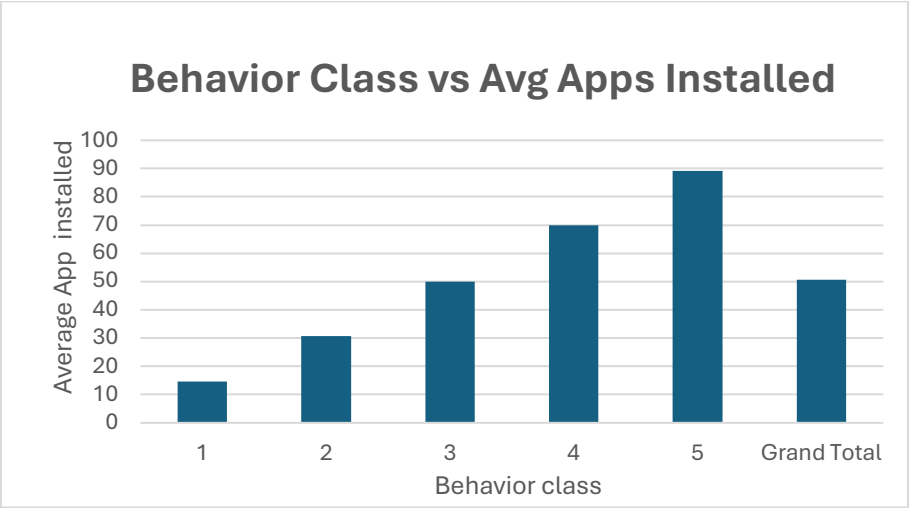
3. Do people in different behavior classes (1–5) behave differently in app usage, screen time, data use, and battery drain?

- **Behavior Class vs App Usage**



From the chart, we can see that the time of app usage generally rises as the behavior class rises. The users in Class 1 use the least time on apps, whereas the users in Class 5 use the maximum time on apps. This is logical, as the behavior classes represent the users, with Class 5 users being the heaviest users of apps.

- **Behavior Class vs Avg Apps Installed**

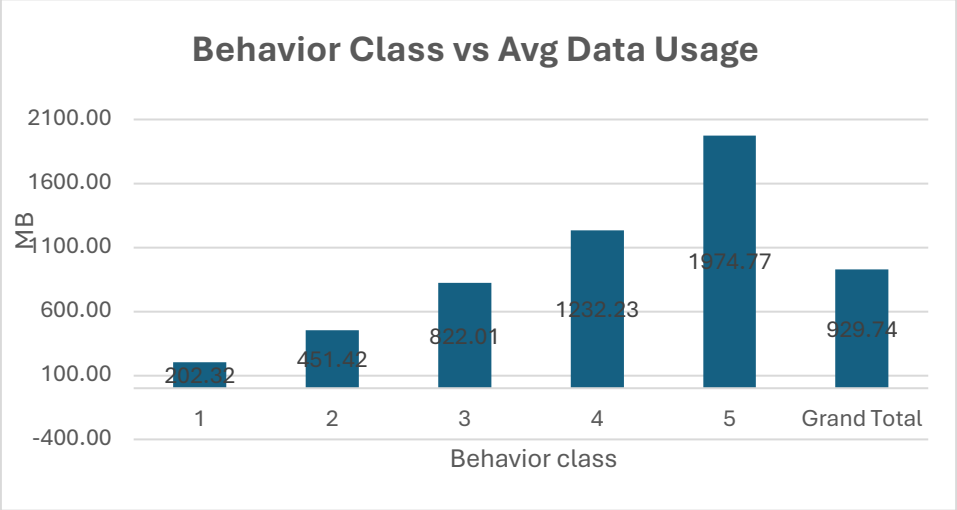


The pattern for screen on time is similar. Lower classes have shorter screen time, and higher classes spend more time actively using their phones. This clearly shows that heavier users keep their screens on for longer periods.

- **Behavior Class vs Avg Screen Time**

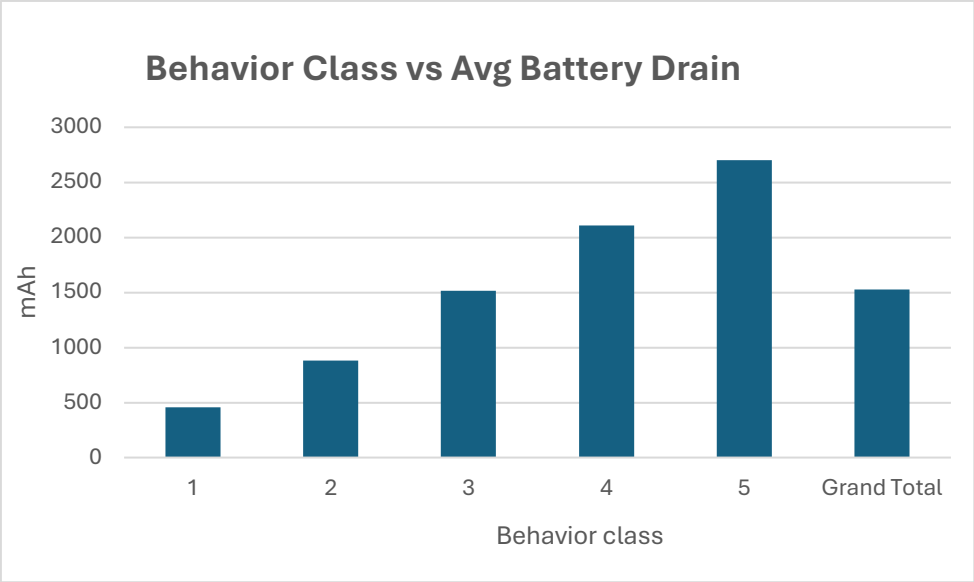
Data usage also rises as we move from Class 1 to Class 5. Users in higher classes consume more mobile data, which likely means they use streaming apps, social media, or online services more frequently. The difference between lower and higher classes shows that behavior class is strongly connected to internet usage.

- **Behavior Class vs Data Usage**



Battery drain increases along with behavior class. This is expected because people who use their phones more will naturally drain the battery faster. Class 5 users show the highest battery consumption, which supports the idea that they are heavy users.

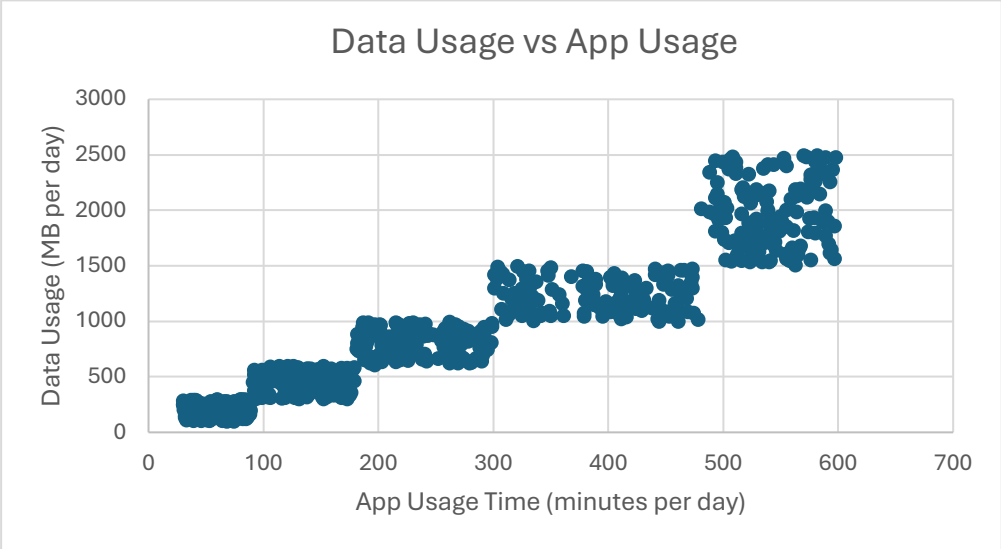
- **Behavior Class vs Avg Battery Drain**



The number of installed apps tends to increase with higher behavior classes. Users who have more apps installed are likely more engaged with their devices. However, the difference between classes may not be very large, which suggests that the number of apps alone does not fully explain user behavior.

4. When one variable increases, does the other also increase or decrease?

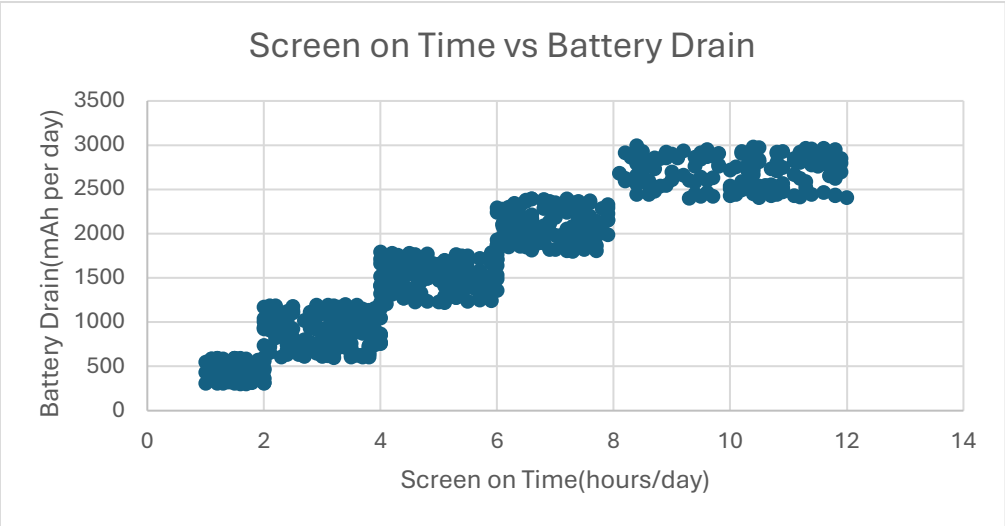
i. App usage time vs Data usage



The scatter plot shows a positive relationship between app usage time and data usage. As app usage time increases, data usage also increases. This indicates that users who spend more time on apps tend to consume more mobile data.

Verdict: App usage time and data usage have a positive relationship, where higher app usage leads to higher data consumption.

ii. Screen on Time vs Battery Drain



The scatter plot indicates a positive relationship between screen on time and battery drain. Users who keep their screens on for longer periods tend to experience higher battery drain, that is a common and expected usage pattern.

Verdict:. App usage time has a positive relationship with data usage, meaning higher app usage leads to higher data consumption. Screen on time also shows a positive relationship with battery drain, where longer screen usage results in greater battery loss